

Qualification Pack



IoT Solution Engineer

QP Code: SSC/Q8202

Version: 4.0

NSQF Level: 6

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Qualification Pack

Contents

SSC/Q8202: IoT Solution Engineer	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SSC/N8246: Design new solution architectures as per specifications for IoT	6
SSC/N8247: Maintain existing architectures & continuously evaluate and upgrade the architecture of IoT solutions	11
SSC/N8210: Design and implement security architecture across end-to-end IoT solutions	17
SSC/N8214: Integrate Information Technology (IT) systems with Operational Technology (OT) systems	22
SSC/N8215: Test and validate architecture of IoT solutions	27
SSC/N8217: Use appropriate standards and ensure compliance of IoT solutions	31
SSC/N8220: Integrate IoT solutions across various device components, networks, databases, analytics solutions and cloud infrastructure	35
SSC/N8248: Design hardware for IoT solutions	39
DGT/VSQ/N0102: Employability Skills (60 Hours)	43
Assessment Guidelines and Weightage	50
<i>Assessment Guidelines</i>	50
<i>Assessment Weightage</i>	50
Acronyms	52
Glossary	53

Qualification Pack

SSC/Q8202: IoT Solution Engineer

Brief Job Description

The IoT Solution Architect role encompasses designing, maintaining, and securing end-to-end IoT solutions. They analyze business needs, select appropriate platforms, and develop system architectures that integrate hardware, software, and communication protocols. They plan for data flow, quality checks, security architecture, and system interoperability, often incorporating AI, edge computing, and cloud services. They evaluate existing architectures for performance, security, reliability, and scalability, ensuring continuous improvement. Additionally, they design hardware components, oversee integration of IT and OT systems, enforce standards compliance, and validate solution effectiveness through testing. Security is prioritized across all layers, with ongoing updates to safeguard against threats. Overall, the architect ensures solutions meet technical, operational, and business objectives effectively and securely.

Personal Attributes

Individuals in this role must possess good interpersonal skills to interact with multiple stakeholders. They must have a strong analytical mindset and showcase a zeal to learn new things.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8246: Design new solution architectures as per specifications for IoT](#)
2. [SSC/N8247: Maintain existing architectures & continuously evaluate and upgrade the architecture of IoT solutions](#)
3. [SSC/N8210: Design and implement security architecture across end-to-end IoT solutions](#)
4. [SSC/N8214: Integrate Information Technology \(IT\) systems with Operational Technology \(OT\) systems](#)
5. [SSC/N8215: Test and validate architecture of IoT solutions](#)
6. [SSC/N8217: Use appropriate standards and ensure compliance of IoT solutions](#)
7. [SSC/N8220: Integrate IoT solutions across various device components, networks, databases, analytics solutions and cloud infrastructure](#)
8. [SSC/N8248: Design hardware for IoT solutions](#)
9. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
Country	India
NSQF Level	6
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	Completed 4 year UG program (PHD or PG or UG or diploma with courses related to Engg./ Science) OR Completed 3 year UG degree (PHD or PG or UG or diploma with courses related to Engg./ Science) with 1.5 years of experience in relevant field. Relevant Experience in job roles related to Computer Science/IT/IoT/ Solution architecture (AI/ML or software development). The relevant experience would include work, internship and apprenticeship after completing relevant educational qualifications. OR Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience in relevant field. Relevant Experience in job roles related to Computer Science/IT/IoT/ Solution architecture (AI/ML or software development). The relevant experience would include work, internship and apprenticeship after completing relevant educational qualifications
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-06-IT-045962025-V2-NASSCOM

Qualification Pack

NQR Version	2
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Qualification Pack

SSC/N8246: Design new solution architectures as per specifications for IoT

Description

This unit is about defining and describing an architecture of a system delivered in context of a specific solution

Scope

The scope covers the following :

- Design IoT Solution architecture as per specifications

Elements and Performance Criteria

Design new solution architectures as per specifications for IoT

To be competent, the user/individual on the job must be able to:

- PC1.** Identify business requirements, analyze Business Objectives and User Needs
- PC2.** Evaluate the platform on which the solution is to be built
- PC3.** Design data feeds, data quality checks, and components needed to configure the solution
- PC4.** Define & plan system-level architecture for the solution
- PC5.** Select appropriate design standards, methods and tools
- PC6.** Select Communication Protocols and Cloud Platforms
- PC7.** Design Security Architecture
- PC8.** Specify Hardware Components and Network Topology
- PC9.** Develop Software and Data Processing Modules
- PC10.** Plan Integration with Existing IT and OT Systems
- PC11.** Conduct Testing and Validation
- PC12.** Iterate and Refine the Design
- PC13.** Prepare Deployment and Monitoring Strategies
- PC14.** Implement Deployment and Maintenance Procedures
- PC15.** Applicability of AI and Machine Learning for IoT Solutions
- PC16.** Applicability of Edge Computing in IoT Architectures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing new solution architectures
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting solution architectures

Qualification Pack

- KU4.** who to involve when designing new solution architectures
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different business requirements
- KU7.** various trends and industry standards on new approaches to solution design
- KU8.** the business and domain in which the solution is to be developed
- KU9.** various governance issues and standards for data storage, security, privacy, and monitoring
- KU10.** different platforms such as mobile application, mobile browser, web browser
- KU11.** different elements of solution architecture such as business architecture, information architecture, application architecture, technology architecture, etc.
- KU12.** various standards for architecture design
- KU13.** how to define technical capabilities roadmap
- KU14.** how to define components and component specifications for solutions
- KU15.** how to identify different quality metrics to monitor the performance of architecture
- KU16.** how to scale the solution architecture
- KU17.** how to explore various cloud or distributed computing platforms such as AWS, Azure, Hadoop, their affiliated services, and how to use these
- KU18.** various networking and hardware systems and their impact on the performance of the solution architecture
- KU19.** how to query and write scripts to acquire and understand the data
- KU20.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable courses
- GS2.** follow rule-based decision-making processes
- GS3.** configure data and disseminate relevant information to others
- GS4.** analyze data and activities
- GS5.** apply problem-solving approaches to different situations
- GS6.** apply balanced judgments to different situations
- GS7.** check the work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design new solution architectures as per specifications for IoT</i>	30	50	-	20
PC1. Identify business requirements, analyze Business Objectives and User Needs	2	3	-	1
PC2. Evaluate the platform on which the solution is to be built	2	3	-	2
PC3. Design data feeds, data quality checks, and components needed to configure the solution	2	3	-	1
PC4. Define & plan system-level architecture for the solution	2	3	-	1
PC5. Select appropriate design standards, methods and tools	2	4	-	2
PC6. Select Communication Protocols and Cloud Platforms	2	3	-	1
PC7. Design Security Architecture	2	3	-	1
PC8. Specify Hardware Components and Network Topology	1	3	-	1
PC9. Develop Software and Data Processing Modules	3	3	-	1
PC10. Plan Integration with Existing IT and OT Systems	3	4	-	2
PC11. Conduct Testing and Validation	1	3	-	1
PC12. Iterate and Refine the Design	1	3	-	1
PC13. Prepare Deployment and Monitoring Strategies	3	3	-	2
PC14. Implement Deployment and Maintenance Procedures	2	3	-	1
PC15. Applicability of AI and Machine Learning for IoT Solutions	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Applicability of Edge Computing in IoT Architectures	1	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8246
NOS Name	Design new solution architectures as per specifications for IoT
Sector	IT-ITeS
Sub-Sector	
Occupation	Internet of Things
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8247: Maintain existing architectures & continuously evaluate and upgrade the architecture of IoT solutions

Description

This unit is about maintaining existing & updtng the architecture for IoT solutions

Scope

The scope covers the following :

- Review and optimize architecture, Monitor standards

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** review existing system architecture designs to ensure balance of functional, service quality, and systems management requirements
- PC2.** review existing system architecture designs to ensure selection of appropriate technology, efficient use of resources, and integration of multiple systems
- PC3.** review technical capabilities roadmap for existing solution architectures
- PC4.** monitor changes in solution design standards and update architecture
- PC5.** update documentation on existing solution architectures for appropriate people
- PC6.** assess IoT architecture to highlight their characteristics, strengths, and weaknesses
- PC7.** evaluate performance based on the responsiveness of the architecture
- PC8.** evaluate the reliability of the architecture based on its ability to keep operating over time
- PC9.** evaluate the availability of the architecture based on the proportion of time the system is up and running
- PC10.** evaluate the security of the architecture based on its ability to resist unauthorized attempts at usage
- PC11.** evaluate the modifiability of architecture in terms of the speed and costs needed to make changes to it
- PC12.** evaluate the portability of the architecture based on its ability to run under different computing environments
- PC13.** perform regular upgrades to IoT architecture
- PC14.** ensure upgrades address increased scalability, reliability, and security demands

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to maintaining existing solution architectures
- KU2.** organizational policies and procedures for sharing data

Qualification Pack

- KU3.** organizational policies and procedures for documenting solution architectures
- KU4.** who to involve when maintaining existing solution architectures
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** organizational policies, procedures, and guidelines that relate to evaluating and upgrading architecture
- KU7.** organizational policies and procedures for sharing data
- KU8.** organizational policies and procedures for documenting architecture performance
- KU9.** who to involve while performing various architecture performance tests and implementing architecture upgrades
- KU10.** the range of standard templates and tools available and how to use them
- KU11.** different business requirements
- KU12.** various trends and industry standards on new approaches to solution design
- KU13.** the business and domain in which the solution is to be developed
- KU14.** various governance issues and standards for data storage, security, privacy, and monitoring
- KU15.** different platforms such as mobile application, mobile browser, web browser
- KU16.** different elements of solution architecture such as business architecture, information architecture, application architecture, technology architecture, etc.
- KU17.** various standards for architecture design
- KU18.** how to define technical capabilities roadmap
- KU19.** how to define components and component specifications for solutions
- KU20.** how to identify different quality metrics to monitor the performance of architecture
- KU21.** how to scale the solution architecture
- KU22.** how to explore various cloud or distributed computing platforms such as AWS, Azure, Hadoop, their affiliated services, and how to use these
- KU23.** various networking and hardware systems and their impact on the performance of the solution architecture
- KU24.** how to query and write scripts to acquire and understand the data
- KU25.** how to work on various operating systems such as Linux, Ubuntu, or Windows
- KU26.** various sources of information available for designing tests and how to access these
- KU27.** difference between different types of testing such as unit, sub-system, system, automated, functional, non-functional
- KU28.** issues that may occur with the testing requirements and how to address these
- KU29.** principles of designing tests
- KU30.** different simulation tool providers such as Autodesk and how to use their tools
- KU31.** how to evaluate and select the best simulation tool for your purposes
- KU32.** different test automation tools such as Selenium and how to use them
- KU33.** how to create suitable tests for particular applications
- KU34.** how to check whether test cases are suitable to be automated
- KU35.** how to access, create and modify different types of: a. high-level scenarios b. test cases c. automatic scripts d. test data e. test plans
- KU36.** different simulation methodologies and procedures used to check the tests are working and are fit for purpose

Qualification Pack

- KU37.** how to analyze and use feedback to improve your simulations
- KU38.** scripting/programming of languages to follow test cases
- KU39.** the current practice followed in the design of tests for software and hardware systems
- KU40.** how to identify and refer anomalies
- KU41.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable courses
- GS2.** follow rule-based decision-making processes
- GS3.** configure data and disseminate relevant information to others
- GS4.** analyze data and activities
- GS5.** apply problem-solving approaches to different situations
- GS6.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. review existing system architecture designs to ensure balance of functional, service quality, and systems management requirements	2	4	-	2
PC2. review existing system architecture designs to ensure selection of appropriate technology, efficient use of resources, and integration of multiple systems	2	4	-	2
PC3. review technical capabilities roadmap for existing solution architectures	2	3	-	1
PC4. monitor changes in solution design standards and update architecture	2	3	-	1
PC5. update documentation on existing solution architectures for appropriate people	2	3	-	1
PC6. assess IoT architecture to highlight their characteristics, strengths, and weaknesses	3	4	-	1
PC7. evaluate performance based on the responsiveness of the architecture	2	3	-	1
PC8. evaluate the reliability of the architecture based on its ability to keep operating over time	2	3	-	1
PC9. evaluate the availability of the architecture based on the proportion of time the system is up and running	2	3	-	1
PC10. evaluate the security of the architecture based on its ability to resist unauthorized attempts at usage	3	4	-	1
PC11. evaluate the modifiability of architecture in terms of the speed and costs needed to make changes to it	2	4	-	2
PC12. evaluate the portability of the architecture based on its ability to run under different computing environments	2	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. perform regular upgrades to IoT architecture	2	4	-	2
PC14. ensure upgrades address increased scalability, reliability, and security demands	2	4	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8247
NOS Name	Maintain existing architectures & continuously evaluate and upgrade the architecture of IoT solutions
Sector	IT-ITeS
Sub-Sector	
Occupation	Internet of Things
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8210: Design and implement security architecture across end-to-end IoT solutions

Description

This unit is about identifying the security requirements of the solution and subsequently planning and designing robust security architectures

Scope

The scope covers the following :

- Design security architecture, Ensure network Security, Validate security architecture, Define security policies and standards, Upgrade security systems

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** acquire a deep understanding of all technology systems of the organization
- PC2.** identify business and security requirements of the solution
- PC3.** identify and adopt security standards and protocols appropriate for the solution
- PC4.** plan, research, and design robust security architectures for the solution
- PC5.** implement security architecture across all layers of the solution (such as the device, network, and cloud layers)
- PC6.** adopt security safeguards such as routers, firewalls, and Identity Access Management tools
- PC7.** protect the IoT network from unauthorized access or malicious intent
- PC8.** ensure only authorized devices should be able to connect to the network
- PC9.** test security architecture to ensure expected behaviour as per requirements
- PC10.** design and implement security policies and standards
- PC11.** ensure that solution architecture implements, operates, and controls prescribed security policies and standards
- PC12.** ensure that security policies and standards are followed at all times
- PC13.** ensure that solution architecture performs day-to-day security processes such as threat and vulnerability management
- PC14.** continuously update and upgrade security systems as needed

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing security architectures
- KU2.** organizational policies and procedures for sharing data

Qualification Pack

- KU3.** organizational policies and procedures for documenting security architectures, authorization protocols, and security policies
- KU4.** who to involve while adopting security safeguards, testing security architecture, and drafting security policies
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different security requirements of the solution and the organization
- KU7.** various trends and industry standards on new approaches to solution design
- KU8.** security requirements of the domain in which the solution is to be developed
- KU9.** different governance issues and standards for data storage, security, privacy, and monitoring
- KU10.** security requirements of various platforms such as mobile application, mobile browser, web browser
- KU11.** various standards for architecture design
- KU12.** how to define a technical capabilities roadmap
- KU13.** how to define components and component specifications for solutions
- KU14.** various quality metrics to monitor the performance of security architecture
- KU15.** how to scale the security architecture
- KU16.** how to explore various cloud or distributed computing platforms such as AWS, Azure, Hadoop, their affiliated services, and how to use these
- KU17.** various networking and hardware systems and their impact on the performance of the solution architecture
- KU18.** how to query and write scripts to acquire and understand the data
- KU19.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** plan and organize the work to achieve targets and meet deadlines
- GS5.** apply problem-solving approaches to different situations
- GS6.** analyze the business impact and disseminate relevant information to others
- GS7.** apply balanced judgments to different situations
- GS8.** check the work is complete and free from errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. acquire a deep understanding of all technology systems of the organization	8	10	-	3
PC2. identify business and security requirements of the solution	3	6	-	2
PC3. identify and adopt security standards and protocols appropriate for the solution	2	2	-	1
PC4. plan, research, and design robust security architectures for the solution	2	2	-	1
PC5. implement security architecture across all layers of the solution (such as the device, network, and cloud layers)	1	3	-	2
PC6. adopt security safeguards such as routers, firewalls, and Identity Access Management tools	3	6	-	2
PC7. protect the IoT network from unauthorized access or malicious intent	1	3	-	1
PC8. ensure only authorized devices should be able to connect to the network	1	3	-	2
PC9. test security architecture to ensure expected behaviour as per requirements	1	3	-	1
PC10. design and implement security policies and standards	2	2	-	1
PC11. ensure that solution architecture implements, operates, and controls prescribed security policies and standards	2	2	-	1
PC12. ensure that security policies and standards are followed at all times	2	2	-	1
PC13. ensure that solution architecture performs day-to-day security processes such as threat and vulnerability management	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. continuously update and upgrade security systems as needed	1	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8210
NOS Name	Design and implement security architecture across end-to-end IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	4
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8214: Integrate Information Technology (IT) systems with Operational Technology (OT) systems

Description

This unit is about performing integrations across IT and OT systems

Scope

The scope covers the following :

- Identify the impact areas and stakeholders , Plan the IT-OT integration, Execute IT-OT integration, Maintain integrated systems

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify defined areas where IT-OT integration can deliver tangible value
- PC2.** identify and engage with stakeholders who will be involved in IT-OT integration
- PC3.** define the scope of the IT-OT integration initiative
- PC4.** assess costs and benefits of performing ITOT integration
- PC5.** evaluate risks, consequences and pitfalls of performing IT-OT integration
- PC6.** document the value of IT-OT integration including potential growth factors such as enabling new business models
- PC7.** assess the need to consult external vendors to perform IT-OT integration
- PC8.** obtain executive buy-in to perform IT-OT integration
- PC9.** evaluate and document current state across people, processes, and technology systems before the IT-OT integration
- PC10.** evaluate and document future state across people, processes, and technology systems post-IT-OT integration
- PC11.** perform IT-OT integration considering factors such as analysis of standards, data ownership, and security policies
- PC12.** ensure that IT-OT integration does not interfere with daily or mission-critical tasks on both IT and OT systems
- PC13.** perform quality testing for IT-OT integrations
- PC14.** determine frequency to review and update integrated systems
- PC15.** review and update integrated systems over regular intervals

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to performing integrations of various systems

Qualification Pack

- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting costs, benefits, and plans to perform IT-OT integrations
- KU4.** who to involve while obtaining buy-in for IT-OT integrations and implementing them
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** business processes and technical capabilities of OT systems
- KU7.** dependencies and the risks associated with the system integrations
- KU8.** the vulnerability caused as an effect of IT-OT integrations
- KU9.** distributed nature of IT and OT systems and devices
- KU10.** different tools for designing data integrations such as Hive or Kafka
- KU11.** multiple data sources, data integration processes and techniques, and how to use them
- KU12.** different cloud or distributed computing platforms such as AWS, Azure, Hadoop, their affiliated services and how to use these
- KU13.** various networking and hardware systems and their impact on the performance of the data architecture
- KU14.** how to query and write scripts to acquire and understand the data
- KU15.** how to store and retrieve information
- KU16.** how to identify and refer anomalies
- KU17.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors
- GS11.** work independently and collaboratively
- GS12.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify defined areas where IT-OT integration can deliver tangible value	3	5	-	2
PC2. identify and engage with stakeholders who will be involved in IT-OT integration	3	5	-	2
PC3. define the scope of the IT-OT integration initiative	2	1	-	1
PC4. assess costs and benefits of performing ITOT integration	2	1	-	1
PC5. evaluate risks, consequences and pitfalls of performing IT-OT integration	3	5	-	2
PC6. document the value of IT-OT integration including potential growth factors such as enabling new business models	2	3	-	1
PC7. assess the need to consult external vendors to perform IT-OT integration	2	3	-	1
PC8. obtain executive buy-in to perform IT-OT integration	3	5	-	2
PC9. evaluate and document current state across people, processes, and technology systems before the IT-OT integration	1	3	-	1
PC10. evaluate and document future state across people, processes, and technology systems post-IT-OT integration	1	3	-	1
PC11. perform IT-OT integration considering factors such as analysis of standards, data ownership, and security policies	3	5	-	2
PC12. ensure that IT-OT integration does not interfere with daily or mission-critical tasks on both IT and OT systems	1	3	-	1
PC13. perform quality testing for IT-OT integrations	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. determine frequency to review and update integrated systems	2	2	-	1
PC15. review and update integrated systems over regular intervals	1	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8214
NOS Name	Integrate Information Technology (IT) systems with Operational Technology (OT) systems
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	1
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8215: Test and validate architecture of IoT solutions

Description

This unit is about performing various tests to validate architecture

Scope

The scope covers the following :

- Carry out usability testing, Carry out performance testing, Carry out compatibility testing, Carry out regression testing, Carry out reliability and scalability testing, Carry out simulation testing

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** test usability of IoT architecture with real users
- PC2.** test user's ease to use the application, flexibility in handling controls, and system's ability to meet its objectives
- PC3.** test performance issues in the design of IoT architecture
- PC4.**
 - test multiple OS versions, browser types, generations of devices, communications nodes, etc.
 - on IoT architecture
- PC5.** perform regression testing to overcome upgrade related issues
- PC6.** use virtualization tools and technologies to perform reliability and scalability testing
- PC7.** build IoT test environments that involve simulation of edge devices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to architecture testing
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting results of architecture validations
- KU4.** who to involve while performing various architecture validations
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** various sources of information available for designing tests and how to access these
- KU7.** difference between different types of testing such as unit, sub-system, system, automated, functional, non-functional
- KU8.** issues that may occur with the testing requirements and how to address these
- KU9.** principles of designing tests
- KU10.** different simulation tool providers such as Autodesk and how to use their tools
- KU11.** how to evaluate and select the best simulation tool for your purposes
- KU12.** different test automation tools such as Selenium and how to use them
- KU13.** how to create suitable tests for particular applications

Qualification Pack

- KU14.** how to check whether test cases are suitable to be automated
- KU15.** how to access, create and modify different types of: a. high-level scenarios b. test cases c. automatic scripts d. test data e. test plans
- KU16.** different simulation methodologies and procedures used to check the tests are working and are fit for purpose
- KU17.** how to analyze and use feedback to improve your simulations
- KU18.** scripting/programming of languages to follow test cases
- KU19.** the current practice followed in the design of tests for software and hardware systems
- KU20.** how to identify and refer anomalies
- KU21.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making process
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors
- GS11.** work independently and collaboratively
- GS12.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. test usability of IoT architecture with real users	4	7	-	2
PC2. test user's ease to use the application, flexibility in handling controls, and system's ability to meet its objectives	4	7	-	3
PC3. test performance issues in the design of IoT architecture	4	7	-	3
PC4. • test multiple OS versions, browser types, generations of devices, communications nodes, etc. • on IoT architecture	4	7	-	3
PC5. perform regression testing to overcome upgrade related issues	4	7	-	3
PC6. use virtualization tools and technologies to perform reliability and scalability testing	5	7	-	3
PC7. build IoT test environments that involve simulation of edge devices	5	8	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8215
NOS Name	Test and validate architecture of IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8217: Use appropriate standards and ensure compliance of IoT solutions

Description

This unit is about ensuring the use of appropriate standards while designing IoT architecture and ensuring that solutions are compliant with regulations

Scope

The scope covers the following :

- Identify standards
- Ensure compliance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** evaluate what value a given standard brings to IoT architecture
- PC2.** evaluate risks of not using a particular standard for the architecture
- PC3.** identify standards that best fit the needs of the architecture
- PC4.** ensure solutions are compliant with industry, technological and organizational regulations
- PC5.** understand risk and compliance requirements that must be followed and document them
- PC6.** identify plan of action to proactively manage risks and offset costs of compliance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to evaluating standards and compliances
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting various standards
- KU4.** who to involve while evaluating internal and external regulations and creating plans to manage risks
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different standards for solution and architecture designs
- KU7.** the requirements of the IoT architecture
- KU8.** internal and external regulations
- KU9.** costs associated with compliance

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making process
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors
- GS11.** work independently and collaboratively
- GS12.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. evaluate what value a given standard brings to IoT architecture	5	8	-	3
PC2. evaluate risks of not using a particular standard for the architecture	5	8	-	3
PC3. identify standards that best fit the needs of the architecture	5	8	-	3
PC4. ensure solutions are compliant with industry, technological and organizational regulations	5	8	-	3
PC5. understand risk and compliance requirements that must be followed and document them	5	9	-	4
PC6. identify plan of action to proactively manage risks and offset costs of compliance	5	9	-	4
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8217
NOS Name	Use appropriate standards and ensure compliance of IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	1
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8220: Integrate IoT solutions across various device components, networks, databases, analytics solutions and cloud infrastructure

Description

This unit is about integrating the IoT solution across its various layers and ensuring the reliability, scalability, and availability of the architecture.

Scope

The scope covers the following :

- Define integration strategy
- Manage integration
- Drive standardization

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify the short-term and long-term integration needs of the solution
- PC2.** implement integrations of services such as data integration, cloud service integration, business intelligence, etc. depending on needs of the solution
- PC3.** create and coordinate roles, responsibilities, and delivery of integration needs across various teams, both internal and external
- PC4.** perform integration across various solution components such as devices, networks, databases and cloud infrastructure
- PC5.** manage architecture design, security architecture design and hardware design across all components of the solution during integration
- PC6.** ensure reliability, scalability, availability of integrated architecture
- PC7.** perform IoT standardization to streamline overall solution integration of IoT projects
- PC8.** ensure that standardization enables interoperability across devices
- PC9.** ensure that standardization leads to reduction in overall project costs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to coordinate integration activities
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting integration requirements and delivery procedures
- KU4.** who to involve while performing integration activities and ensuring the reliability of integrated architecture
- KU5.** the range of standard templates and tools available and how to use them

Qualification Pack

- KU6.** technical features of the various components of a solution
- KU7.** the dependencies and the risks associated with performing integrations
- KU8.** the vulnerability caused as an effect of integrations
- KU9.** the distributed nature of various systems and devices
- KU10.** different tools for designing data integrations such as Hive or Kafka
- KU11.** different data sources, data integration processes and techniques and how to use them
- KU12.** various cloud or distributed computing platforms such as AWS, Azure, Hadoop, their affiliated services and how to use these
- KU13.** different networking and hardware systems and their impact on the performance of the data architecture
- KU14.** how to query and write scripts to acquire and understand the data
- KU15.** how to store and retrieve information
- KU16.** how to identify and refer anomalies
- KU17.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making process
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** provide opinions on work in a detailed and constructive way
- GS10.** apply balanced judgments to different situations
- GS11.** check the work is complete and free from errors
- GS12.** work independently and collaboratively
- GS13.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify the short-term and long-term integration needs of the solution	3	5	-	2
PC2. implement integrations of services such as data integration, cloud service integration, business intelligence, etc. depending on needs of the solution	3	5	-	2
PC3. create and coordinate roles, responsibilities, and delivery of integration needs across various teams, both internal and external	4	9	-	3
PC4. perform integration across various solution components such as devices, networks, databases and cloud infrastructure	6	9	-	3
PC5. manage architecture design, security architecture design and hardware design across all components of the solution during integration	3	5	-	2
PC6. ensure reliability, scalability, availability of integrated architecture	3	5	-	2
PC7. perform IoT standardization to streamline overall solution integration of IoT projects	3	5	-	2
PC8. ensure that standardization enables interoperability across devices	3	5	-	2
PC9. ensure that standardization leads to reduction in overall project costs	2	2	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8220
NOS Name	Integrate IoT solutions across various device components, networks, databases, analytics solutions and cloud infrastructure
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	6
Credits	1
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8248: Design hardware for IoT solutions

Description

This unit is about driving the hardware development process including designing, assembling and transferring hardware designs to manufacturing teams

Scope

The scope covers the following :

- Design device hardware, Validate certifications and compliance, Ensure hardware security

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** design and assemble all hardware solution components
- PC2.** perform build vs buy analysis of hardware components (such as control systems, enclosures and actuators)
- PC3.** develop designs for components to be built internally
- PC4.** transfer designs to manufacturing teams and provide them with appropriate technical support
- PC5.** perform hardware component selection for components to be procured externally
- PC6.** liaise with multiple external hardware partners and evaluate partners based on required specifications
- PC7.** create and maintain device catalogue
- PC8.** ensure field readiness of devices and tolerance to environmental conditions
- PC9.** ensure certification requirements are met for all hardware components used including flameproof / safety / regulatory requirements
- PC10.** perform pre-compliance testing for hardware components
- PC11.** implement security measures to protect against tampering, IP theft, reverse engineering attacks, etc
- PC12.** ensure hardware security measures against Differential Power Analysis (DPA) attacks

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing, assembling, and manufacturing solution components
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting component designs and specifications
- KU4.** who to involve while developing and transferring designs

Qualification Pack

- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different hardware design options and how to select the best option to meet the requirements and constraints
- KU7.** the production and manufacturing capabilities of the organization
- KU8.** the production and manufacturing capabilities of external vendor partners
- KU9.** how to ensure physical readiness and security of devices
- KU10.** various component certifications
- KU11.** production level scripting
- KU12.** techniques and tools to simulate, analyze and synthesize design options
- KU13.** types of hardware description languages & tools and how to use them, such as Verilog, VHDL, Field Programmable Gate Arrays (FPGA), DVT (Design, Validation, Test), firmware fundamentals, etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** provide opinions on work in a detailed and constructive way
- GS10.** apply balanced judgments to different situations
- GS11.** check the work is complete and free from errors
- GS12.** work independently and collaboratively
- GS13.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. design and assemble all hardware solution components	5	7	-	2
PC2. perform build vs buy analysis of hardware components (such as control systems, enclosures and actuators)	3	4	-	2
PC3. develop designs for components to be built internally	4	6	-	2
PC4. transfer designs to manufacturing teams and provide them with appropriate technical support	2	3	-	2
PC5. perform hardware component selection for components to be procured externally	2	4	-	2
PC6. liaise with multiple external hardware partners and evaluate partners based on required specifications	2	4	-	2
PC7. create and maintain device catalogue	2	2	-	1
PC8. ensure field readiness of devices and tolerance to environmental conditions	2	4	-	2
PC9. ensure certification requirements are met for all hardware components used including flameproof / safety / regulatory requirements	2	4	-	2
PC10. perform pre-compliance testing for hardware components	2	4	-	1
PC11. implement security measures to protect against tampering, IP theft, reverse engineering attacks, etc	2	4	-	1
PC12. ensure hardware security measures against Differential Power Analysis (DPA) attacks	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8248
NOS Name	Design hardware for IoT solutions
Sector	IT-ITeS
Sub-Sector	
Occupation	Internet of Things
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8246.Design new solution architectures as per specifications for IoT	30	50	-	20	100	12
SSC/N8247.Maintain existing architectures & continuously evaluate and upgrade the architecture of IoT solutions	30	50	-	20	100	12
SSC/N8210.Design and implement security architecture across end-to-end IoT solutions	30	50	-	20	100	12
SSC/N8214.Integrate Information Technology (IT) systems with Operational Technology (OT) systems	30	50	-	20	100	12
SSC/N8215.Test and validate architecture of IoT solutions	30	50	-	20	100	12
SSC/N8217.Use appropriate standards and ensure compliance of IoT solutions	30	50	-	20	100	12
SSC/N8220.Integrate IoT solutions across various device components, networks, databases, analytics solutions and cloud infrastructure	30	50	-	20	100	12
SSC/N8248.Design hardware for IoT solutions	30	50	-	20	100	12
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	4
Total	260	430	-	160	850	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.